Siddharth Gurumurthy

Curriculum Vitae

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Education

2017 - 2020 **Bachelor of Science (Honours), Mathematics and Computer Science**, *Chennai Mathematical Institute*
Grade : 8.82/10

2020 - 2021 **Master of Science, Mathematics**,
University of Western Ontario,
Advisor: Dan Christensen
Average grade: 94.12/100

2021 - 2027 (Expected) **PhD, Mathematics**, *University of Rochester*, Advisor: Doug Ravenel

Selected Conferences and Workshops Attended

May 2019 **AIS Homotopy Theory (Workshop)**, ISI Kolkata

Feb 2020 **NCM Workshop on Global Homotopy Theory**, IIT Bombay

May 2022 **Algebraic structures in topology**, San Juan, Puerto Rico

July 2022 **Homotopical Methods in Fixed Point Theory**, University of Colorado, Boulder

Nov 2022–25 **BUGCAT**, Binghamton University

March 2023 **Homotopy theory in honor of Paul Goerss**, Northwestern University

June 2023 **Scissors Congruence, Algebraic K-Theory, and Trace Methods**, Indiana University

Dec 2024 **Life After the Telescope Conjecture**, SLMath

May 2024 **New horizons for equivariance in homotopy theory**, *Isaac Newton Institute*, Cambridge

Selected Expository Talks

Nov 2019 **Categorical Semantics for Intuitionistic Logic**

Dec 2020 **Elementary Topoi**

Mar 2021 **On Hilbert's 17th problem using Model Theory**

June 2021 **The Joyal Model Structure**, *Western University*, Homotopy Theory seminar

April 2022 **Elliptic Curves, Formal Groups and Cohomology Theories**

Sep 2022 **Stable Homotopy Theory: History and Motivation**, *University at Buffalo*, Graduate Seminar

March 2023 **Stacks in Homotopy Theory**, *University of Rochester*, Topology Seminar

April 2023 **Galois Theory and Covering Spaces**, *University of Rochester*, Graduate Seminar

Sep 2023 **On the Hopf Invariant One problem**, *University at Buffalo*, Graduate Seminar

Dec 2023 **Combinatorial Species: Investigating the zoo of structures**, *University of Rochester*, Graduate Seminar

Dec 2025 **Formal Groups and Cohomology Theories**, *University of Rochester*, Graduate Seminar

Research Talks

Nov 2025 **Thom spectra revisited**, *University of Minnesota*, Topology Seminar

Nov 2025 **Thom spectra revisited**, *Binghamton University*, BUGCAT Conference

Teaching Experience

Teaching Assistant

Spring 2019 **APRG**, *Advanced Programming in Python*, Chennai Mathematical Institute

Spring 2020 **MATH 1120B**, *Fundamental Concepts of Mathematics*, Western University

Fall 2021 **MATH 265**, *Functions of a Real Variable*, University of Rochester

Spring 2022 **MATH 235**, *Linear Algebra*, University of Rochester

Fall 2022 **MATH 141**, *Calculus I*, University of Rochester

Spring 2023 **MATH 162**, *Calculus IIA*, University of Rochester

Summer 2023 **MATH 120**, *Precalculus Module*, University of Rochester

Fall 2023 **MATH 201**, *Introduction to Probability*, University of Rochester

Spring 2024 **MATH 164**, *Multidimensional Calculus*, University of Rochester

Fall 2024 **MATH 440**, *Graduate Topology*, University of Rochester

Spring 2025 **MATH 240H**, *Introduction to Topology, Honors*, University of Rochester

Fall 2025 **MATH 440**, *Graduate Topology*, University of Rochester

Spring 2026 **MATH 162**, *Calculus IIA*, University of Rochester

Fall 2026 **MATH 440**, *Graduate Topology*, University of Rochester

Preprints

1. In Progress: **Monoidality of the Parametrized Straightening-Unstraightening Construction** joint work with Ryan Quinn and Qi Zhu. See the research statement for more details.

Service

2023-2025 **Rochester Graduate Student Seminar**,
Organized the graduate student seminar in the math department

Reading Projects for Undergraduates

Fall 2021 **Universal Algebra and Category Theory**

Fall 2022–24 **Introduction to Category Theory**

Experience with Programming Languages

Mostly in the context of competitive programming, but also some real- world experience: C++, Python, Haskell.